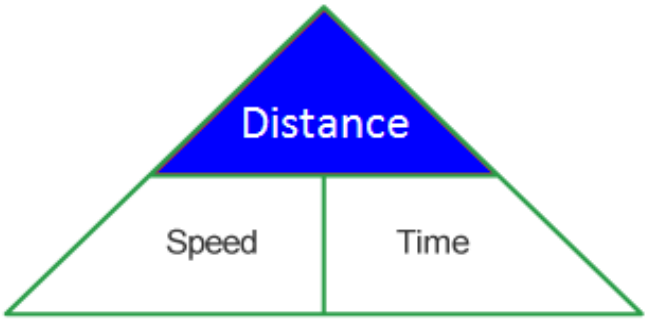
Online Session Date:6/8/2020

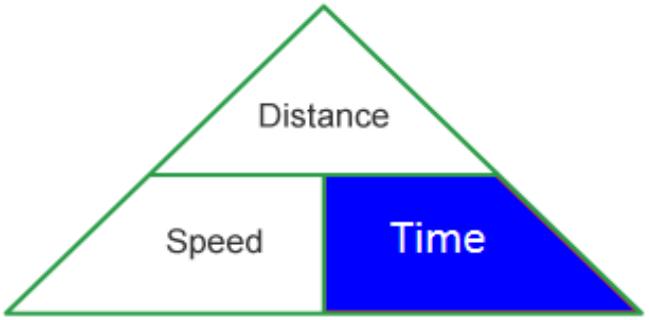
Unit- II

Time and Distance, Problem on Trains, Problem on Boat

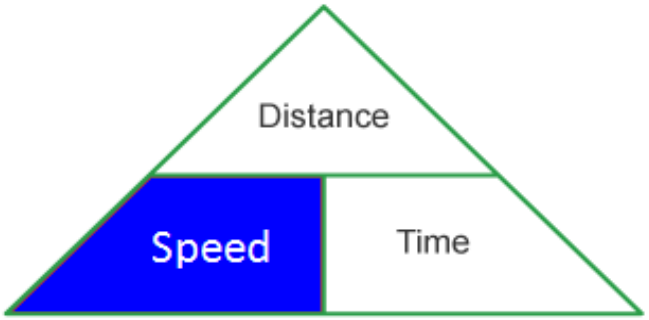
Time and Distance



$$\text{Distance} = \text{Speed} \times \text{Time}$$



$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$



$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Time and Distance

Important Facts and Formulae:

1. Speed $= \left(\frac{\text{Distance}}{\text{Time}}\right)$, Time $= \left(\frac{\text{Distance}}{\text{Speed}}\right)$, Distance = (Speed × Time)

2. x km/hr $= \left(x \times \frac{5}{18}\right)$ m/sec

3. x m/sec $= \left(x \times \frac{18}{5}\right)$ km/hr

4. If the ratio of the speeds of A and B is $a : b$, then the ratio of the times taken by them to cover the same distance is $\frac{1}{a} : \frac{1}{b}$ or $b : a$.

5. Suppose a man covers a certain distance at x km/hr and an equal distance at y km/hr. Then, the average speed during the whole journey is $\left(\frac{2xy}{x+y}\right)$ km/hr.

Ex. 1. ***How many minutes does Aditya take to cover a distance of 400 m, if he runs at a speed of 20 km/hr ?*** **(Bank P.O. 2000)**

Sol. Aditya's speed = 20 km/hr $= \left(20 \times \frac{5}{18}\right)$ m/sec $= \frac{50}{9}$ m/sec.

$\therefore$ Time taken to cover 400 m $= \left(400 \times \frac{9}{50}\right)$ sec = 72 sec $= 1\frac{12}{60}$ min $= 1\frac{1}{5}$ min.

Ex. 2. ***A cyclist covers a distance of 750 m in 2 min 30 sec. What is the speed in km/hr of the cyclist ?*** **(R.R.B. 2002)**

Sol. Speed $= \left(\frac{750}{150}\right)$ m/sec = 5m/sec $= \left(5 \times \frac{18}{5}\right)$ km/hr = 18 km/hr.

. Mark (✓) against the correct answer :

1. A car moves at the speed of 80 km / hr. What is the speed of the car in metres per second ? **(Hotel Management. 2002)**

(a) 8 m / sec (b) $20\frac{1}{9}$ m / sec (c) $22\frac{2}{9}$ m / sec (d) None of these

2. An athlete runs 200 metres race in 24 seconds. His speed is : **(S.S.C. 2002)**

(a) 20 km / hr (b) 24 km / hr (c) 28.5 km / hr (d) 30 km / hr

3. Which of the following trains is the fastest ?

(a) 25 m / sec (b) 1500 m / min (c) 90 km / hr (d) None of these

4. A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour ?

(a) 3.6 (b) 7.2 (c) 8.4 (d) 10

1. Speed $= \left(80 \times \frac{5}{18}\right)$ m/sec $= \frac{200}{9}$ m/sec $= 22\frac{2}{9}$ m/sec.

2. Speed $= \frac{200}{24}$ m/sec $= \frac{25}{3}$ m/sec $= \left(\frac{25}{3} \times \frac{18}{5}\right)$ km/hr $= 30$ km/hr.

3. 25 m/sec $= \left(25 \times \frac{18}{5}\right)$ km/hr $= 90$ km/hr.

   And, 25 m/sec $= (25 \times 60)$ m/min $= 1500$ m/min.

   So, all the three speeds are equal.

4. Speed $= \left(\frac{600}{5 \times 60}\right)$ m/sec $= 2$ m/sec $= \left(2 \times \frac{18}{5}\right)$ km/hr $= 7.2$ km/hr.

5. A man walking at the rate of 5 km / hr crosses a bridge in 15 minutes. The length of the bridge (in metres) is : (S.S.C. 2000)

(a) 600 (b) 750 (c) 1000 (d) 1250

Solution-speed = (5x5/18)m/sec=25/18m/sec

Distance covered in 15 minutes=25/18x15x60

= 1250m

6. How long will a boy take to run round a square field of side 35 metres, if he runs at the rate of 9 km / hr ? (S.S.C. 1999)

(a) 50 sec (b) 52 sec (c) 54 sec (d) 56 sec

Solution- speed =(9x5/18)=5/2m/sec

Distance = 35x4=140m

Time=140/(5/2)= (140x2/5)=56 sec

Question:- An aeroplane covers a certain distance at a speed of 240 kmph in 5 hours. To cover the same distance in 1 2/3 hours, it must travel at a speed of:

A.300 kmph **B.**360 kmph **C.**600 kmph **D.**720 kmph

Solution- Dsitance =240x5=1200km

speed=D/T= 1200/(5/3)

= 1200x3/5=720 km/hr

Question:- If a person walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. The actual distance travelled by him is:

A.50 km **B.**56 km **C.**70 km **D.**80 km

Solution- The actual distance travelled be x km

$x/10=x+20/14$

$14x=10x+200$

$4x=200$

$X=50km$

Ex. 6. *A man travelled from the village to the post-office at the rate of 25 kmph and walked back at the rate of 4 kmph. If the whole journey took 5 hours 48 minutes, find the distance of the post-office from the village.* (S.S.C. 2004)

Sol. Average speed $= \left(\frac{2xy}{x+y}\right)$ km/hr $= \left(\frac{2 \times 25 \times 4}{25+4}\right)$ km/hr $= \frac{200}{29}$ km/hr.

Distance travelled in 5 hours 48 minutes *i.e.*, $5\frac{4}{5}$ hrs $= \left(\frac{200}{29} \times \frac{29}{5}\right)$ km $= 40$ km.

$\therefore$ Distance of the post-office from the village $= \left(\frac{40}{2}\right) = 20$ km.

Ex. 10. *A and B are two stations 390 km apart. A train starts from A at 10 a.m. and travels towards B at 65 kmph. Another train starts from B at 11 a.m. and travels towards A at 35 kmph. At what time do they meet ?*

Sol. Suppose they meet x hours after 10 a.m. Then,

(Distance moved by first in x hrs) + [Distance moved by second in $(x-1)$ hrs] = 390.

$\therefore \quad 65x + 35(x-1) = 390 \Rightarrow 100x = 425 \Rightarrow x = 4\frac{1}{4}$.

So, they meet 4 hrs. 15 min. after 10 a.m. *i.e.*, at 2.15 p.m.

14. An express train travelled at an average speed of 100 km / hr, stopping for 3 minutes after every 75 km. How long did it take to reach its destination 600 km from the starting point ? (M.A.T. 2003)

(a) 6 hrs 21 min (b) 6 hrs 24 min (c) 6 hrs 27 min (d) 6 hrs 30 min

Solution- T=D/S

Time taken to cover 600km=600/100=6hrs

Number of stoppages = 600/75-1=7

Total time of stoppage=3x7=21min

Total time=6hrs 21 min

- A man complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.

A.220 km **B.**224 km **C.**230 km **D.**234 km

Solution- $(1/2)x/21 + (1/2)x/24 = 10$

$x/21 + x/24 = 20$

$45x/504 = 20$

$45x = 20 \times 504$

$X = 10080/45$ $x = 224$

(I.A.F.

28. A person travels from P to Q at a speed of 40 kmph and returns by increasing his speed by 50%. What is his average speed for both the trips ? (M.B.A. 2003)

(a) 36 kmph (b) 45 kmph (c) 48 kmph (d) 50 kmph

28. Speed on return trip = 150% of 40 = 60 kmph.

$$\therefore \text{ Average speed } = \left(\frac{2 \times 40 \times 60}{40 + 60}\right) \text{km/hr} = \left(\frac{4800}{100}\right) \text{km/hr} = 48 \text{ km/hr.}$$

Train

1. Time taken by a train of length l metres to pass a pole or standing man or a signal post is equal to the time taken by the train to cover l metres.
2. Time taken by a train of length l metres to pass a stationery object of length b metres is the time taken by the train to cover $(l + b)$ metres.
3. Suppose two trains or two objects bodies are moving in the same direction at u m/s and v m/s, where $u > v$, then their relative speed is = $(u - v)$ m/s.
4. Suppose two trains or two objects bodies are moving in opposite directions at u m/s and v m/s, then their relative speed is = $(u + v)$ m/s.
5. If two trains of length a metres and b metres are moving in opposite directions at u m/s and v m/s, then:

   The time taken by the trains to cross each other $= \frac{(a + b)}{(u + v)}$ sec.

6. If two trains of length a metres and b metres are moving in the same direction at u m/s and v m/s, then:

   The time taken by the faster train to cross the slower train $= \frac{(a + b)}{(u - v)}$ sec.

7. If two trains (or bodies) start at the same time from points A and B towards each other and after crossing they take a and b sec in reaching B and A respectively, then:

   (A's speed) : (B's speed) = $(\sqrt{b} : \sqrt{a})$

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1. A train running at the speed of 60 km/hr crosses a pole in 9 seconds. What is the length of the train?
   A.120 metres
   B.180 metres
   C.324 metres
   D.150 metres

2. A train 125 m long passes a man, running at 5 km/hr in the same direction in which the train is going, in 10 seconds. The speed of the train is:
   A.45 km/hr
   B.50 km/hr
   C.54 km/hr

D.55 km/hr
The length of the bridge, which a train 130 metres long and travelling at 45 km/hr can cross in 30 seconds, is:
A.200 m
B.225 m
C.245 m
D.250 m

4. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:
A.1 : 3
B.3 : 2
C.3 : 4
D.None of these

5. A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/hr, what is the length of the platform?
A.120 m
B.240 m
C.300 m
D.None of these

6. A train 240 m long passes a pole in 24 seconds. How long will it take to pass a platform 650 m long?
A.65 sec
B.89 sec
C.100 sec
D.150 sec

7. Two trains of equal length are running on parallel lines in the same direction at 46 km/hr and 36 km/hr. The faster train passes the slower train in 36 seconds. The length of each train is:
A.50 m
B.72 m
C.80 m
D.82 m

8. A train 360 m long is running at a speed of 45 km/hr. In what time will it pass a bridge 140 m long?
A.40 sec
B.42 sec
C.45 sec
D.48 sec

9. Two trains are moving in opposite directions @ 60 km/hr and 90 km/hr. Their lengths are 1.10 km and 0.9 km respectively. The time taken by the slower train to cross the faster train in seconds is:
   A.36
   B.45
   C.48
   D.49

10. A jogger running at 9 kmph alongside a railway track in 240 metres ahead of the engine of a 120 metres long train running at 45 kmph in the same direction. In how much time will the train pass the jogger?
   A.3.6 sec
   B.18 sec
   C.36 sec
   D.72 sec

11. A 270 metres long train running at the speed of 120 kmph crosses another train running in opposite direction at the speed of 80 kmph in 9 seconds. What is the length of the other train?
   A.230 m
   B.240 m
   C.260 m
   D.320 m
   E.None of these

12. A goods train runs at the speed of 72 kmph and crosses a 250 m long platform in 26 seconds. What is the length of the goods train?
   A.230 m
   B.240 m
   C.260 m
   D.270 m

13. Two trains, each 100 m long, moving in opposite directions, cross each other in 8 seconds. If one is moving twice as fast the other, then the speed of the faster train is:
   A.30 km/hr
   B.45 km/hr
   C.60 km/hr
   D.75 km/hr

14. Two trains 140 m and 160 m long run at the speed of 60 km/hr and 40 km/hr respectively in opposite directions on parallel tracks. The time (in seconds) which they take to cross each other, is:
   A.9
   B.9.6

C.10
D.10.8

15. A train 110 metres long is running with a speed of 60 kmph. In what time will it pass a man who is running at 6 kmph in the direction opposite to that in which the train is going?
A.5 sec
B.6 sec
C.7 sec
D.10 sec

16. A train travelling at a speed of 75 mph enters a tunnel 3 $^{1}/_{2}$ miles long. The train is ¼ mile long. How long does it take for the train to pass through the tunnel from the moment the front enters to the moment the rear emerges?
A.2.5 min
B.3 min
C.3.2 min
D.3.5 min

17. A train 800 metres long is running at a speed of 78 km/hr. If it crosses a tunnel in 1 minute, then the length of the tunnel (in meters) is:
A.130
B.360
C.500
D.540

18. A 300 metre long train crosses a platform in 39 seconds while it crosses a signal pole in 18 seconds. What is the length of the platform?
A.320 m
B.350 m
C.650 m
D.Data inadequate

19. A train speeds past a pole in 15 seconds and a platform 100 m long in 25 seconds. Its length is:
A.50 m
B.150 m
C.200 m
D.Data inadequate

20. A train moves past a telegraph post and a bridge 264 m long in 8 seconds and 20 seconds respectively. What is the speed of the train?
A.69.5 km/hr
B.70 km/hr
C.79 km/hr
D.79.2 km/hr

21. How many seconds will a 500 metre long train take to cross a man walking with a speed of 3 km/hr in the direction of the moving train if the speed of the train is 63 km/hr?
A.25
B.30
C.40
D.45

22. Two goods train each 500 m long, are running in opposite directions on parallel tracks. Their speeds are 45 km/hr and 30 km/hr respectively. Find the time taken by the slower train to pass the driver of the faster one.
A.12 sec
B.24 sec
C.48 sec
D.60 sec

23. Two trains are running in opposite directions with the same speed. If the length of each train is 120 metres and they cross each other in 12 seconds, then the speed of each train (in km/hr) is:
A.10
B.18
C.36
D.72

24. Two trains of equal lengths take 10 seconds and 15 seconds respectively to cross a telegraph post. If the length of each train be 120 metres, in what time (in seconds) will they cross each other travelling in opposite direction?
A.10
B.12
C.15
D.20

25. A train 108 m long moving at a speed of 50 km/hr crosses a train 112 m long coming from opposite direction in 6 seconds. The speed of the second train is:
A.48 km/hr
B.54 km/hr
C.66 km/hr
D.82 km/hr

Boat & Stream

1. Downstream/Upstream:

In water, the direction along the stream is called *downstream*. And, the direction against the stream is called *upstream*.

2. If the speed of a boat in still water is u km/hr and the speed of the stream is v km/hr, then:

Speed downstream = $(u + v)$ km/hr.

Speed upstream = $(u - v)$ km/hr.

3. If the speed downstream is a km/hr and the speed upstream is b km/hr, then:

Speed in still water $= \frac{1}{2}(a + b)$ km/hr.

Rate of stream $= \frac{1}{2}(a - b)$ km/hr.

1. A boat can travel with a speed of 13 km/hr in still water. If the speed of the stream is 4 km/hr, find the time taken by the boat to go 68 km downstream.
   A.2 hours
   B.3 hours
   C.4 hours
   D.5 hours

2. A man's speed with the current is 15 km/hr and the speed of the current is 2.5 km/hr. The man's speed against the current is:
   A.8.5 km/hr
   B.9 km/hr
   C.10 km/hr
   D.12.5 km/hr

4. A motorboat, whose speed in 15 km/hr in still water goes 30 km downstream and comes back in a total of 4 hours 30 minutes. The speed of the stream (in km/hr) is:
   A.4
   B.5
   C.6
   D.10

5. In one hour, a boat goes 11 km/hr along the stream and 5 km/hr against the stream. The speed of the boat in still water (in km/hr) is:
   A.3 km/hr
   B.5 km/hr
   C.8 km/hr
   D.9 km/hr

6. A boat running downstream covers a distance of 16 km in 2 hours while for covering the same distance upstream, it takes 4 hours. What is the speed of the boat in still water?
   A.4 km/hr
   B.6 km/hr
   C.8 km/hr
   D.Data inadequate

7. The speed of a boat in still water in 15 km/hr and the rate of current is 3 km/hr. The distance travelled downstream in 12 minutes is:
   A.1.2 km
   B.1.8 km
   C.2.4 km
   D.3.6 km

8. A boat takes 90 minutes less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 mph, the speed of the stream is:
   A.2 mph
   B.2.5 mph
   C.3 mph
   D.4 mph

9. A man can row at 5 kmph in still water. If the velocity of current is 1 kmph and it takes him 1 hour to row to a place and come back, how far is the place?
   A.2.4 km
   B.2.5 km
   C.3 km
   D.3.6 km

10. A boat covers a certain distance downstream in 1 hour, while it comes back in 1 hours. If the speed of the stream be 3 kmph, what is the speed of the boat in still water?

A.12 kmph
B.13 kmph
C.14 kmph
D.15 kmph
E.None of these

11. A person can swim in still water at 4 km/h. If the speed of water 2 km/h, how many hours will the man take to swim back against the current for 6km?

A. 3 B. 4

C. 4 (1/2) D. Insufficient data

12. A boat goes 100 km downstream in 10 hours, and 75 m upstream in 15 hours. The speed of the stream is?

A. 7 km/h B. 5 km/h

C. 3 km/h D. 2 (1/2) km/h

13. A man can row his boat with the stream at 6 km/h and against the stream in 4 km/h. The man's rate is?

A. 1 kmph B. 5 kmph

C. 8 kmph D. 3 kmph

14. A man can swim in still water at 4.5 km/h, but takes twice as long to swim upstream than downstream. The speed of the stream is?

A. 3 B. 7.5

C. 2.25 D. 1.5

15. A man can row with a speed of 15 kmph in still water. If the stream flows at 5 kmph, then the speed in downstream is?

A. 10 kmph B. 5 kmph

C. 20 kmph D. 22 kmph

16. A man can row upstream at 25 kmph and downstream at 35 kmph, and then find the speed of the man in still water?

A. 60 kmph

B. 10 kmph

C. 30 kmph

D. 5 kmph

17. A boat can move upstream at 25 kmph and downstream at 35 kmph, then the speed of the current is?

A. 5 kmph

B. 10 kmph

C. 30 kmph

D. 15 kmph

18. A man swims downstream 30 km and upstream 18 km taking 3 hours each time, what is the speed of the man in still water?

A. 2 kmph

B. 8 kmph

C. 16 kmph

D. 4 kmph

19. A man swims downstream 72 km and upstream 45 km taking 9 hours each time; what is the speed of the current?

A. 3 kmph

B. 1.5 kmph

C. 13 kmph

D. 6.5 kmph

20. A man can row 6 kmph in still water. When the river is running at 1.2 kmph, it takes him 1 hour to row to a place and black. How far is the place?

A. 3.12 km

B. 2.88 km

C. 3 km

D. 2 km

21. A man can row 6 kmph in still water. When the river is running at 1.2 kmph, it takes him 1 hour to row to a place and black. What is the total distance traveled by the man?

A. 6.24 km

B. 6 km

C. 5.76 km

D. 5.66 km

22. The current of a stream runs at the rate of 4 kmph. A boat goes 6 km and back to the starting point in 2 hours, then find the speed of the boat in still water?

A. 10 kmph

B. 21 kmph

C. 8 kmph

D. 12 kmph

23. A man whose speed is 4.5 kmph in still water rows to a certain upstream point and back to the starting point in a river which flows at 1.5 kmph, find his average speed for the total journey?

A. 8 kmph

B. 4 kmph

C. 2 kmph

D. 10 kmph

24. The current of a stream at 1 kmph. A motor boat goes 35 km upstream and back to the starting point in 12 hours. The speed of the motor boat in still water is?

A. 6 kmph

B. 7 kmph

C. 8.5 kmph

D. 8 kmph

25. The speed of a boat in still water is 60kmph and the speed of the current is 20kmph. Find the speed downstream and upstream?

A. 35, 25 kmph
B. 80, 40 kmph
C. 40, 60 kmph
D. 50, 55 kmph